

Shading

Computer Graphics
Instructor: Sungkil Lee

Today

- **Overview**
- **Light source and materials**
- **Phong illumination model**
- **Blinn-Phong illumination model**
- **Polygonal shading**

Shading

- **Shading:**

- is a process of altering the color of an object/surface/polygon in a 3D scene, based on how lights and materials interact.

light와 material의 interaction에 따라 color를 바꿔주는 것
(오프젝트와 물리적인 model에 따라 그려주는 것)



VS.



Color or albedo

그냥 색깔만

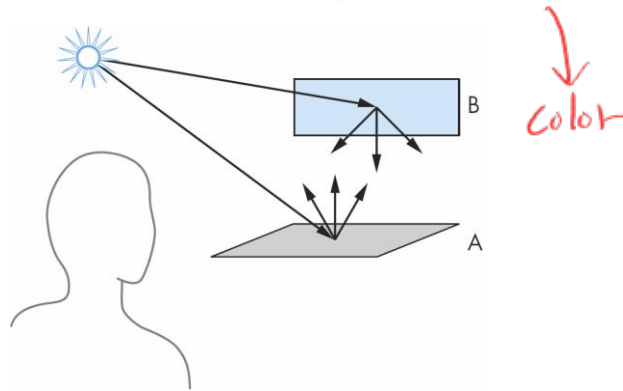
Shading

Light-Material Interaction

- **Light-material interactions**

- cause each point to have a different color or shade.
- Light reaching surfaces are partially absorbed, and some reflected.

surface에 닿은 빛이 흡수되거나 반사됨



- **The color (or brightness) of an object**

- The amounts of reflected lights
- For example, a surface appears red under white light, because the red component of the light is reflected and the rest absorbed.

Light Sources and Materials

Elements of Light-Material Interaction

- **Four elements on which light reflection depends:**

- **Light sources**
- **Surface materials** (e.g., smoothness) of the object surfaces
 - Material indicates actually an internal atomic/molecular structure.
- **Surface orientation:** indicated by normal vectors
- **Location of viewer:** eye/camera position in our viewing

표면이 거칠거나 매끄러운지, 반사되는 각도가 어떻게 달라지는지에만 관심

Abstract Modeling

- **Abstract modeling**

abstraction

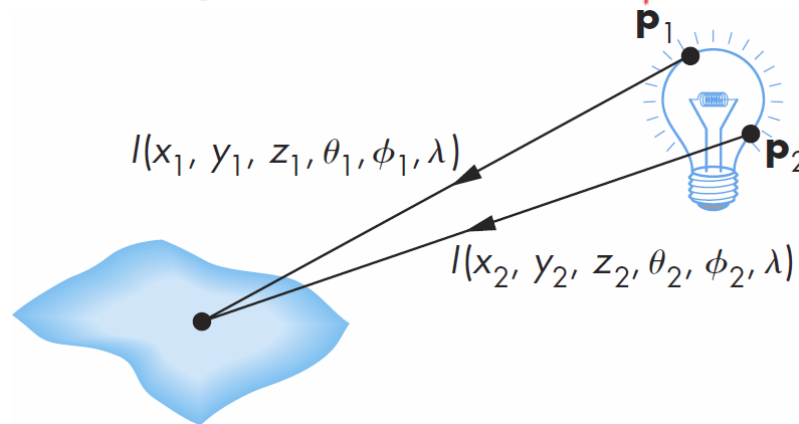
- Light sources and surface materials are modeled as abstract forms.
- Location of viewer and surface orientation are used as they are.

- **Example: general lights**

geometry라서 정확함

- are hard to work with, because we must integrate light coming from all points on the source.
- Instead, we consider simpler models, which effectively abstracts general light sources.

general light source를 point로 modeling



Light Source Models

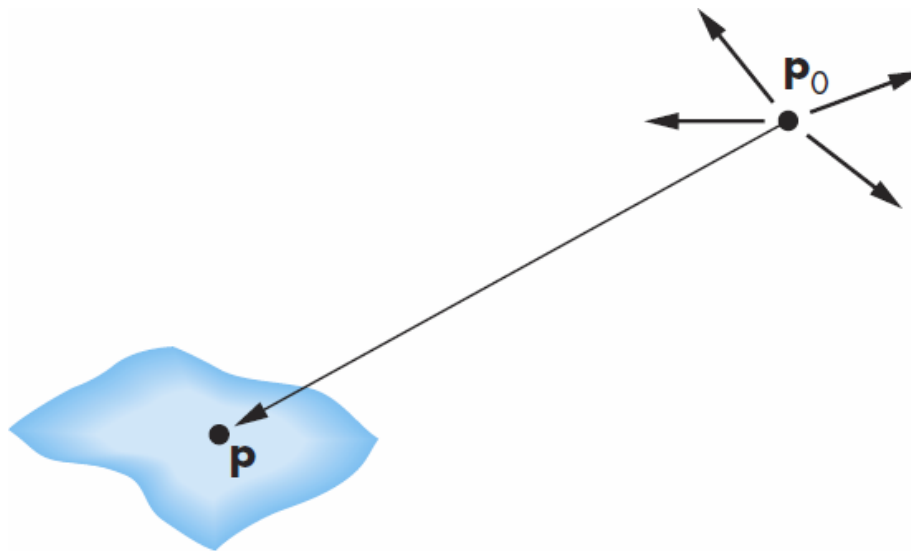
- **Three simplified light source models**
 - Point light source
 - Distant (or directional) light source
 - Ambient light

Light Source Models

- **Point light source**

- Model with position and color.
- The light source position is a point.
- We may apply **distance attenuation**, inversely scaling with the square of distances from a light source to surface points.

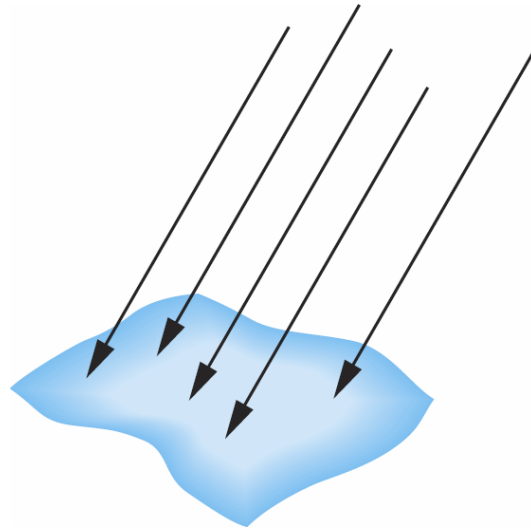
거리가 멀수록 빛의 양이 줄어든다



Light Source Models

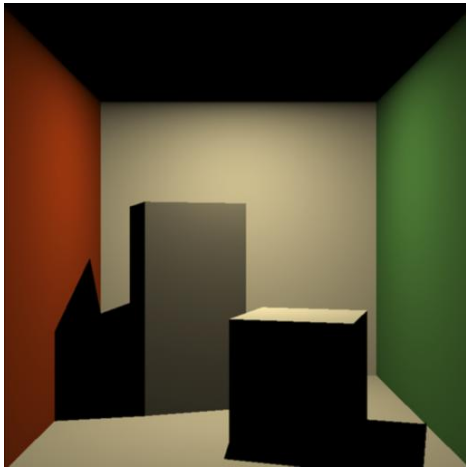
- **Distant (directional) light source**

- Infinite distance away (e.g., parallel sunlight)
- The light source has only a direction vector.
 - In case of HC (i.e., vec4), the last element w is 0.



Light Source Models

- **Ambient light** *Indirect light을 approximation한 것*
 - Same amount of light everywhere in a scene, like a **shading offset**
 - Can model contribution of many sources and indirect reflections.



Direct



Indirect

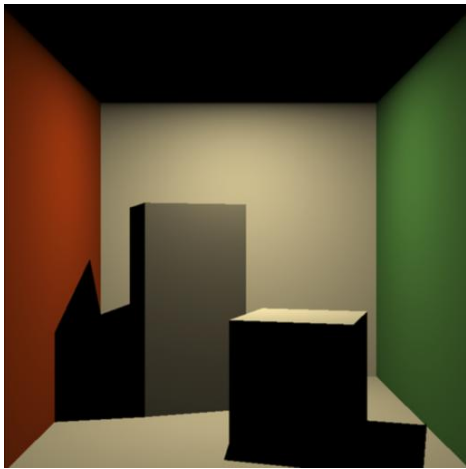


Direct + Indirect
(global shading)

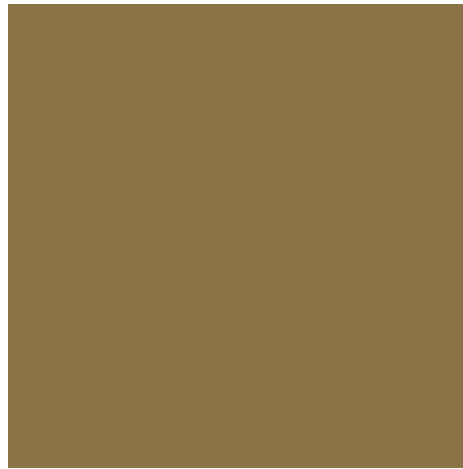
Light Source Models

- **Ambient light**

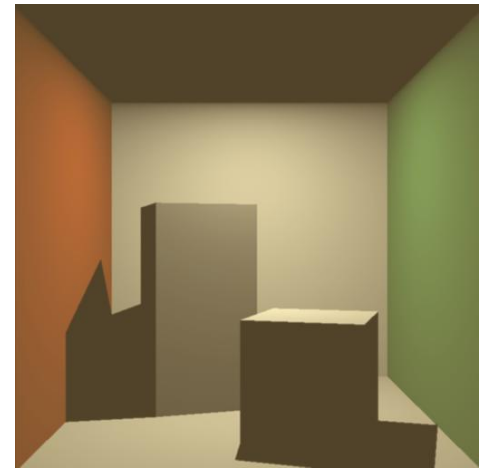
- Same amount of light everywhere in a scene, like a shading offset
- Can model contribution of many sources and indirect reflections.
- This, however, is significant approximation, and limited in practice.



Direct



Ambient
그냥 offset
(상수를 더함)



Direct + Ambient
크게 개선되는 게 없음

Surface Materials

- **Microfacet model**

surface 아주 작은 거울로 이루어짐

- Surface are composed of many very tiny facets, each of which is a perfect specular reflector (i.e., mirror).
- The distribution of the microfacets or their normals determine the characteristics of surface reflection.



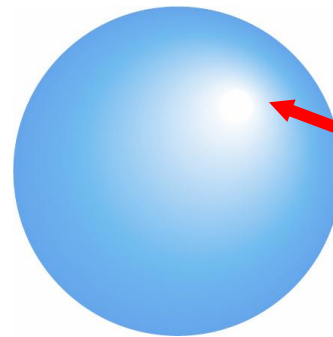
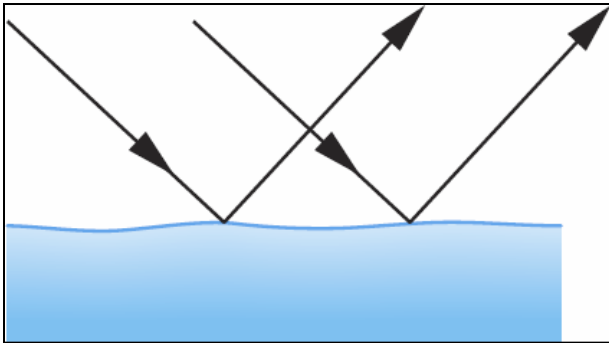
Smooth and rough microfacets

Types of Surface Materials

가을반사 가의 가을반사

- **Specular/glossy surfaces**

- Smooth surfaces show specular highlights. *매끄러워서 잘 반사함*
- The smoother a surface is, the more reflected light is concentrated into the direction of a perfect mirror reflection.



specular highlight

(가을 반사되는 곳)

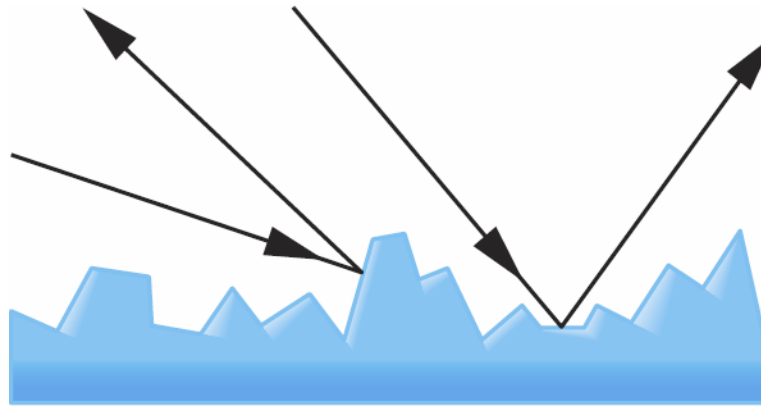
Types of Surface Materials

(대략) 2분

- **Diffuse reflector**

- A very rough surface scatters light equally in all directions.
- Amount of light reflected is proportional to the vertical component of incoming light.
- Ideal (or perfect) diffuse surfaces are called "**Lambertian**" surfaces.

이상적인 반사



Surface Materials

- **The types of surface materials**

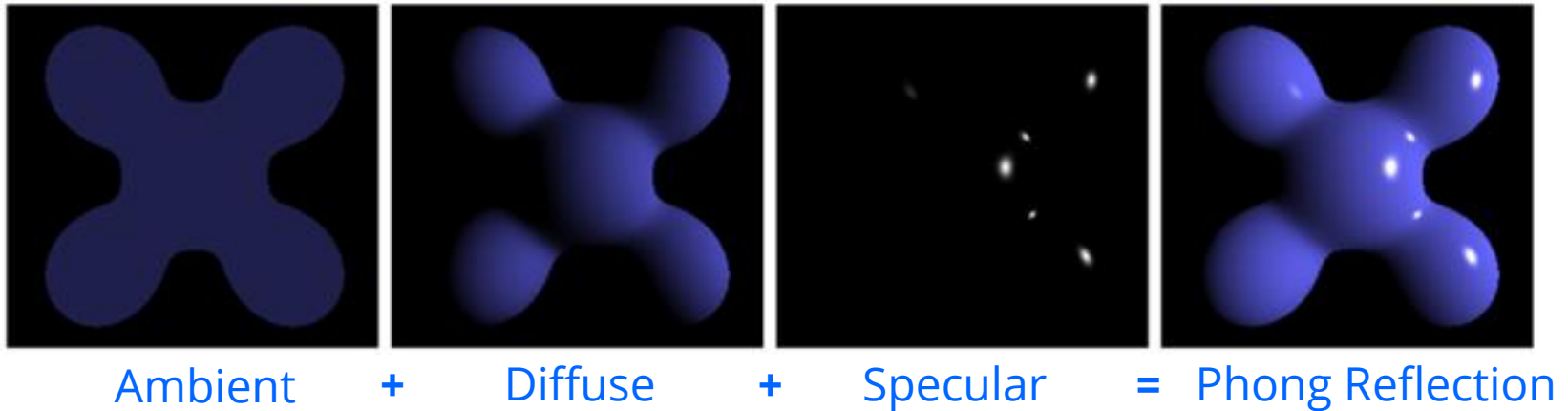
- Most surfaces are neither ideally rough or perfectly smooth.
- Hence, we decompose light source and surface materials, and combine them for the final shading.
- Phong illumination model implements this idea.

Specular surface > 반사율 → R_s
Lambertian

Phong Illumination Model

Phong Illumination Model

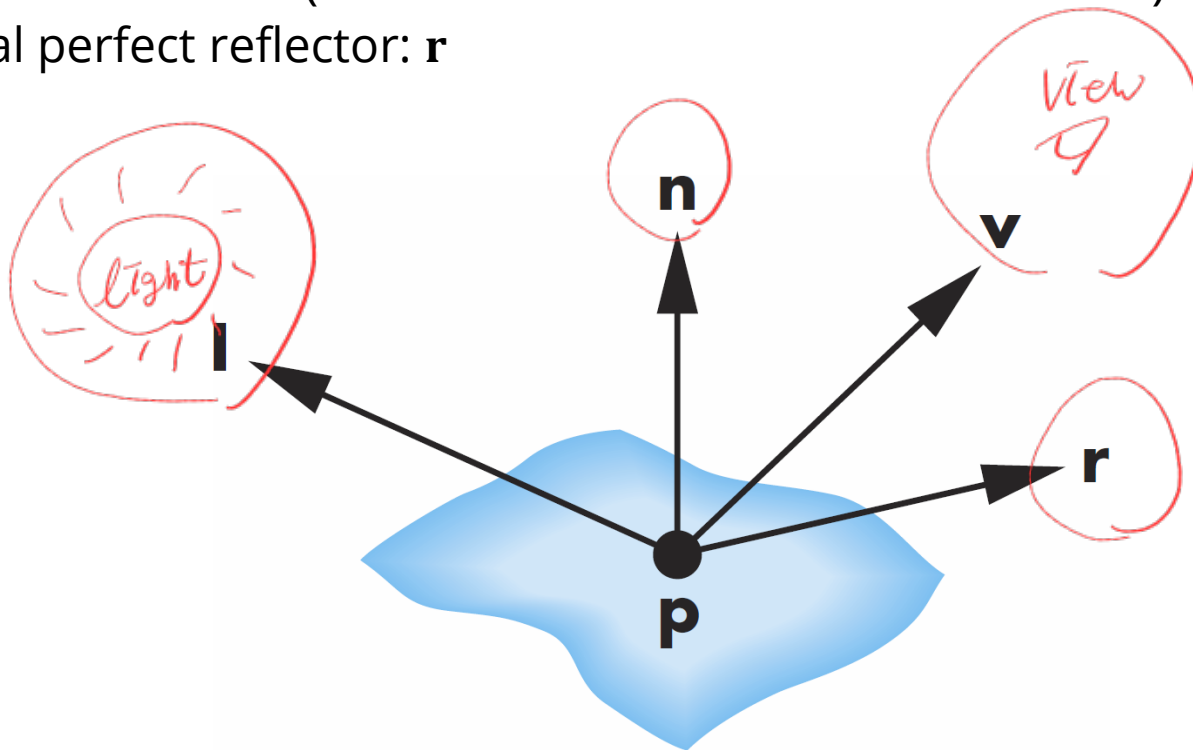
- **A simple empirical model that**
 - can be computed rapidly in the pipeline approach.
 - the form does not have perfect physical justification.
- **Three illumination components:**
 - Ambient, diffuse, and specular



Phong Illumination Model

- **Uses four vectors**

- To light source: $\mathbf{l}$ (a vector ***not from light source but to light source***)
- To viewer: $\mathbf{v}$ (*camera - p* camera space라든 camera가 원점이므로 -p)
- Normal vector: $\mathbf{n}$ (will be discussed later in more details)
- Ideal perfect reflector: $\mathbf{r}$



Perfect Reflector r

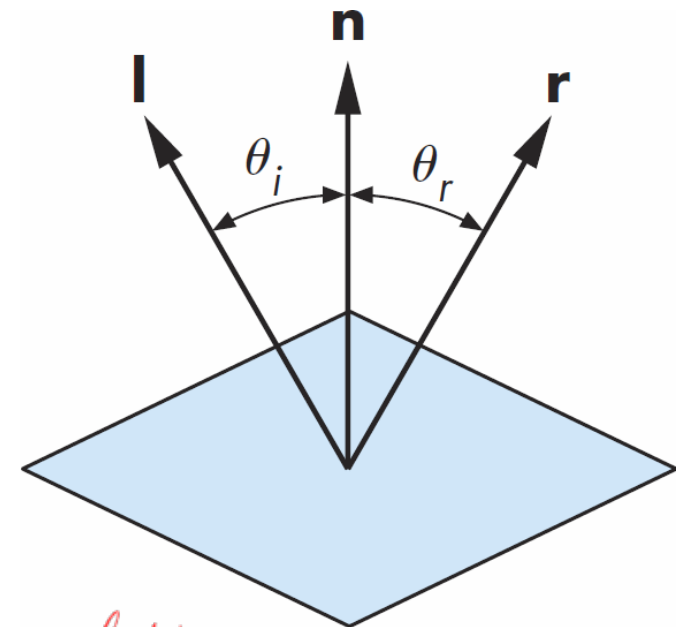
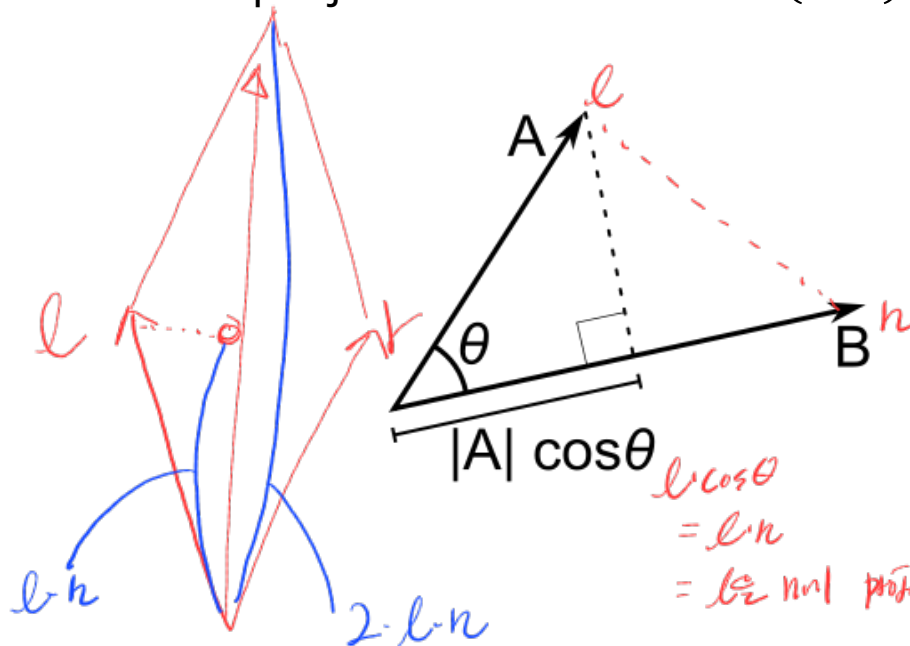
- While l and n are given, r needs to be computed

- Angle of incidence (θ_i) = angle of reflection (θ_r)
- The three vectors must be coplanar.
- The reflector is given as:

$l \cdot n$: scalar

$$\mathbf{r} = 2(\mathbf{l} \cdot \mathbf{n})\mathbf{n} - \mathbf{l}$$

- Note: projection of l onto n is $(\mathbf{l} \cdot \mathbf{n})\mathbf{n}$



$$\begin{aligned} l + r &= s \cdot n \\ r &= s \cdot n - l \\ s &= 2(l \cdot n) \\ r &= 2(l \cdot n) - l \end{aligned}$$

Ambient Reflections

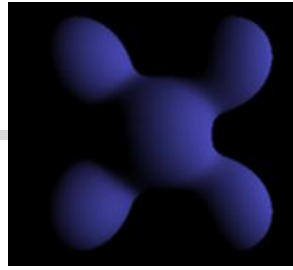


- **Ambient reflections**

- Amounts of reflected lights depend on both the color of the lights and the material properties (i.e., reflectance; k_a) of the object.
- This is a coarse approximation to the secondary (indirect) reflections among the surfaces.

$$I_{ra} = \underbrace{k_a}_{\text{reflectance}} \underbrace{I_a}_{\text{ambient}}$$

Diffuse Reflections



- **Diffuse reflections**

- Similar to ambient reflection, but the angular attenuation is included here:

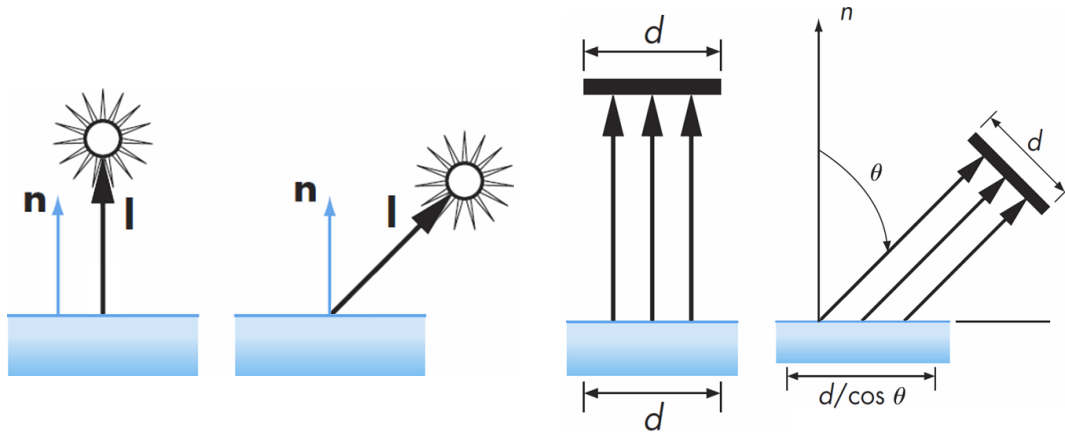
$$I_{rd} = \max((\mathbf{l} \cdot \mathbf{n})k_d I_d, 0)$$

광원과 normal vector가 반대로 맞으면 음수가 나오기 때문에 0으로

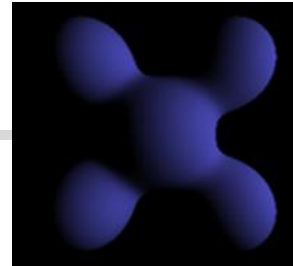
diffuse light의 intensity

- **Angular attenuation** $(\mathbf{l} \cdot \mathbf{n})$: 빛이 들어오는 각도

- amount of incoming light scales with the angle of incoming light.
- reflected light $\approx \cos \theta_i = \mathbf{l} \cdot \mathbf{n}$



Diffuse Reflections



- We can further incorporate **distance attenuation**,

$$I_{rd} = \text{att}(d) \cdot \max(\mathbf{k}_d(\mathbf{l} \cdot \mathbf{n})I_d, 0)$$

- **Distant attenuation**

- The light from a point source that reaches a surface is inversely proportional to the square of the distance (d)

광원에서 멀어질수록 빛이 약해짐

$$\text{att}(d) = \frac{1}{a + bd + cd^2}, \quad d \uparrow \text{ 멀어질수록 작아짐}$$

- a, b, c are user-defined coefficients. 각 값 보편성 맞춰줘야 됨
- The constant and linear terms (a and b) soften the effect of the point source

Specular Reflections

- **Specular reflections (with distance attenuation):**

$$I_{rs} = \text{att}(d) \cdot \max(\mathbf{k}_s I_s (\mathbf{r} \cdot \mathbf{v})^\alpha, 0)$$

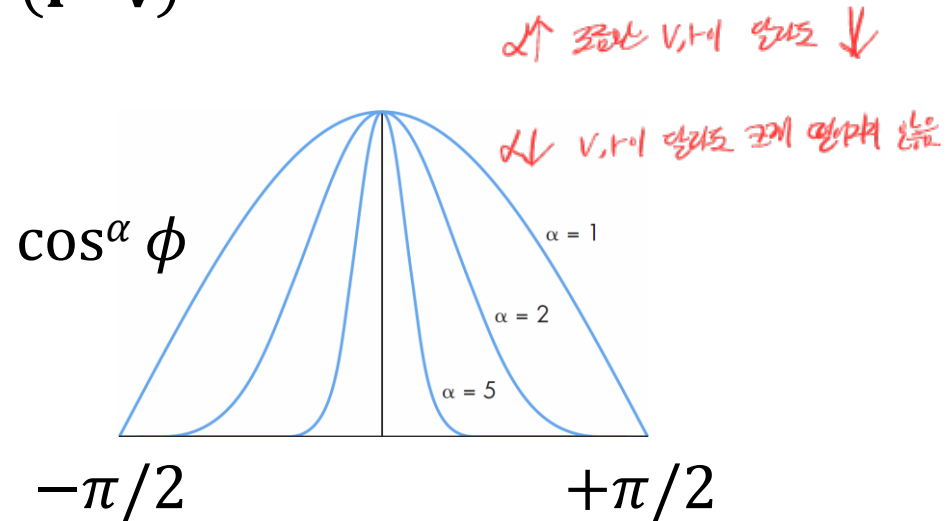
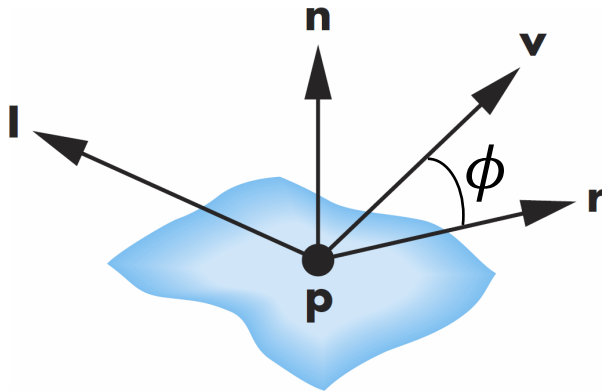
use가 줄

v가 r과 동일한 방향에서 r을 볼 수 있음 (서로 가까운 수록 빛이 세는 것)

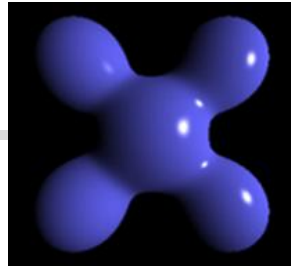
- The ideal reflector $\mathbf{r}$ closer to the viewer direction $\mathbf{v}$ shines more.
- Hence, the angle between $\mathbf{v}$ the $\mathbf{r}$ can be a drop-off term.

줄어들

$$\cos^\alpha \phi = (\mathbf{r} \cdot \mathbf{v})^\alpha$$



Putting them altogether



- **For each light source and each color component,**
 - the Phong model can be written (without distance terms) as:

$$\begin{aligned} I_r &= I_{ra} + I_{rd} + I_{rs} \\ &= \underline{k_a I_a} + \underline{\max(k_d (\mathbf{l} \cdot \mathbf{n}) I_d, 0)} + \underline{\max(k_s (\mathbf{r} \cdot \mathbf{v})^\alpha I_s, 0)} \end{aligned}$$

- You can also include distance attenuation term to the model, if necessary.

$$\begin{aligned} \text{att}(d) \\ &= \frac{1}{a + bd + cd^2} \end{aligned}$$

Blinn-Phong Model: **Efficient approximation to Phong Model**

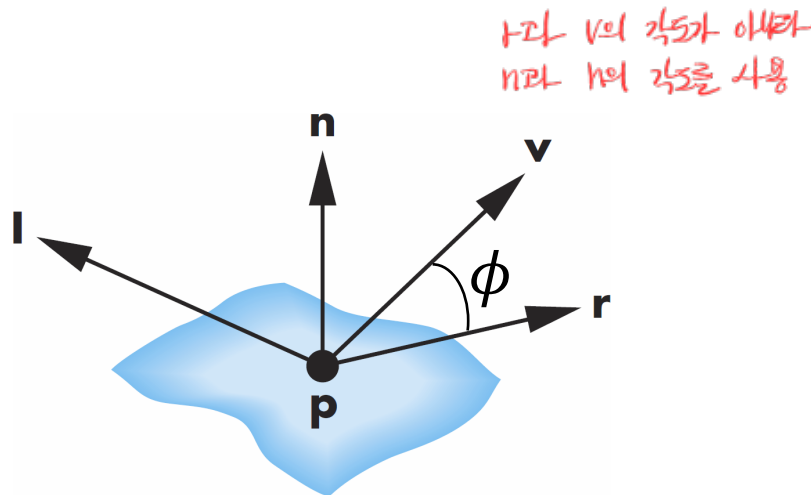
Modified Phong Model

- The **specular term** in Phong model

- calculates $\mathbf{r}$ and $\mathbf{v}$ for each vertex

$$\underline{k_s (\mathbf{r} \cdot \mathbf{v})^\alpha \mathbf{I}_s}$$

- recall that $\mathbf{r} = 2(\mathbf{l} \cdot \mathbf{n})\mathbf{n} - \mathbf{l}$
- Blinn suggested an efficient approximation using the halfway vector.

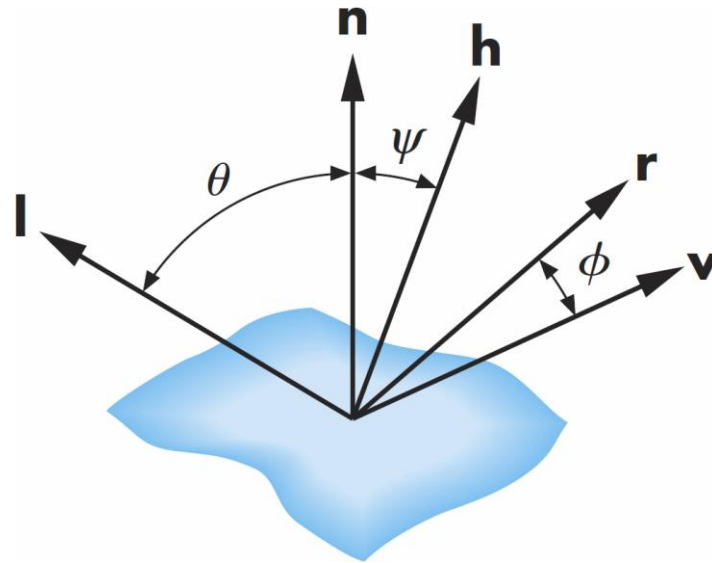


The Halfway Vector

- **The halfway vector $\mathbf{h}$:**

- A normalized vector halfway between $\mathbf{l}$ and $\mathbf{v}$.
- When $\mathbf{v}$ is coplanar with $\mathbf{l}$, $\mathbf{n}$, and $\mathbf{r}$, we can show that $2\psi = \phi$.

$$\mathbf{h} = \frac{\mathbf{l} + \mathbf{v}}{|\mathbf{l} + \mathbf{v}|}$$



Blinn-Phong Model

- The **Blinn-Phong** model can be written:

$$\begin{aligned} I'_r &= I_{ra} + I_{rd} + I'_{rs} \\ &= k_a I_a + \max(k_d (l \cdot n) I_d, 0) + \max(k_s (\mathbf{n} \cdot \mathbf{h})^\beta I_s, 0) \end{aligned}$$

- Replace $(v \cdot r)^\alpha$ with $(n \cdot h)^\beta$
- β is chosen to match shininess α . *α와 비슷하게 보이게 해주는 β를 새로 찾음*
- This model has been a standard in the old-style OpenGL.

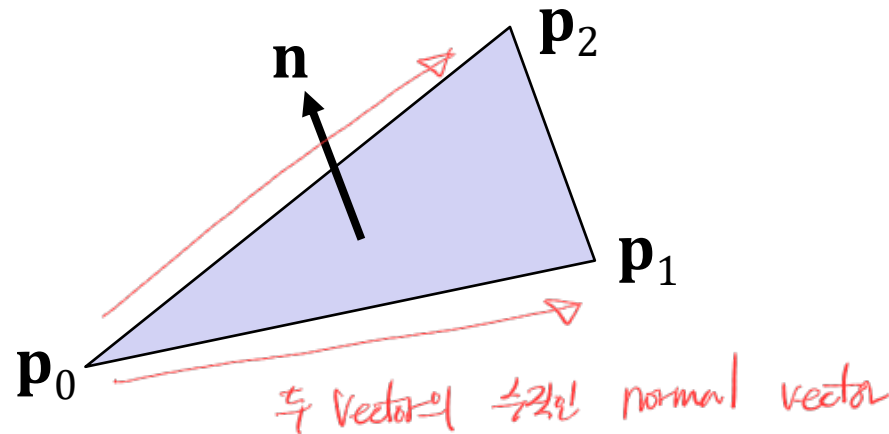
Polygonal Shading: Effects of Normal Vectors

Per-Face Normals

- Normal vectors usually indicate face (or per-face) normals.
 - Given 3 vertices, $\mathbf{p}_0, \mathbf{p}_1, \mathbf{p}_2$, the *surface normal* is given as:

$$\mathbf{n} = (\mathbf{p}_1 - \mathbf{p}_0) \otimes (\mathbf{p}_2 - \mathbf{p}_0)$$

cross product
외적



Discontinuity with Face Normals

- **However, polygonal meshes is composed of discrete polygons (or triangles).**
 - The discontinuity of per-face normals causes the discontinuity of shadings around the face boundaries. For example,

face마다 normal vector가 다름
→ 부드럽게 만들고 싶으면 너무 많은
수를 줄이기 위해 interpolation



- **So, it is better to use different normals:**
 - **which can be interpolated**
 - either of per-vertex normals or per-fragment normals

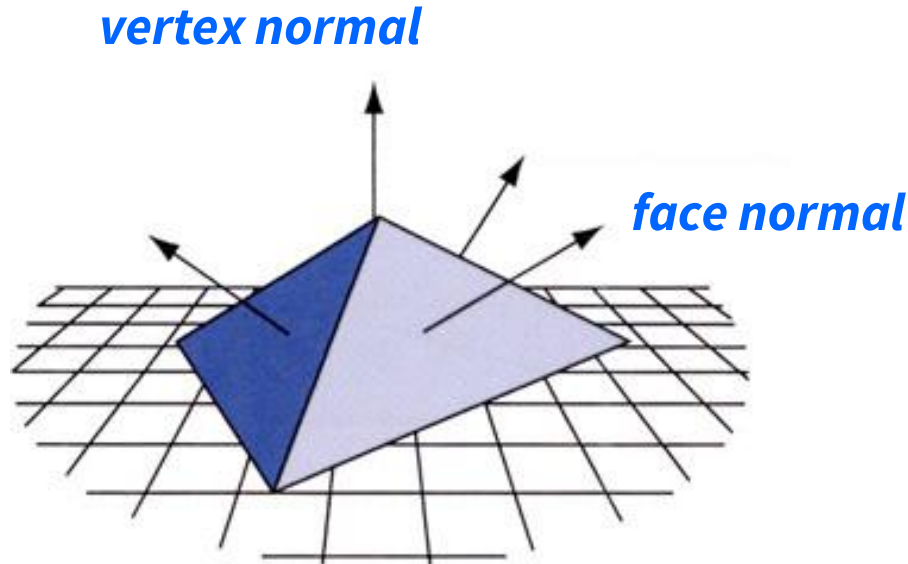
(Per-) Vertex Normals

Vertex는 point, normal vector가 있는게 이상함

인접한 face들의 normal을 평균을 사용

- A **vertex normal** is the average of adjacent face normals.

- Physically, this does not make sense, but provides smoother shading by interpolating the normals of adjacent vertices. 물리적으로는 맞지 않지만 시각적인 결과가 좋음
- Hence, this is the default choice for normals in CG.
 - Saying just "normals" means vertex normals in practice in CG.



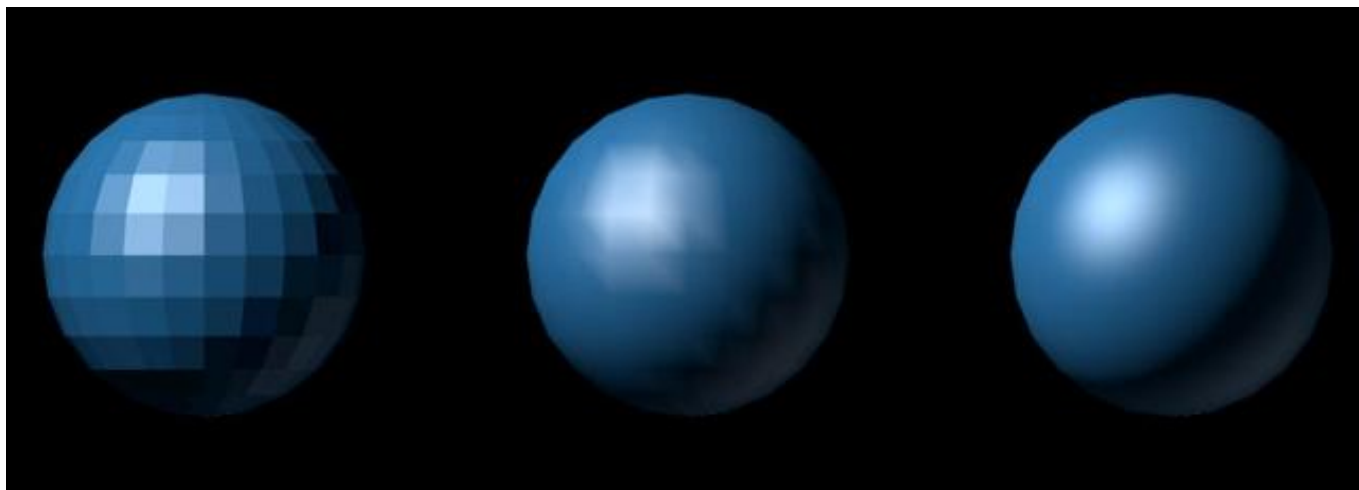
Three Polygonal Shading Methods

- **Shading implementation relies on:**

- normals: either per-face or per-vertex normals
- interpolation: either colors or normals

- **There are three techniques**

- Flat shading (face normal) raster에서 면 interpolation 하지 않음
- Gouraud shading → color
- Phong shading → normalization



Three Polygonal Shading Methods

- **Where shading is applied for each method with normals:**

- Only Phong shading computes in FS, while the others in VS.
- Now, we see why we call vertex "shader."
 - Gouraud shading is computed in vertex shader.
 - It was a standard for fixed OpenGL pipeline (in old-style OpenGL).

Method	Input normals	Vertex shader	Interpolation in Rasterizer		Fragment shader
			from	to	
Flat	face normals	shading			
Gouraud	vertex normals	shading	vertex <u>colors</u>	fragment <u>colors</u>	
Phong	vertex normals		vertex <u>normals</u>	<u>fragment normals</u>	- renormalize fragment normals - shading with renormalized normals

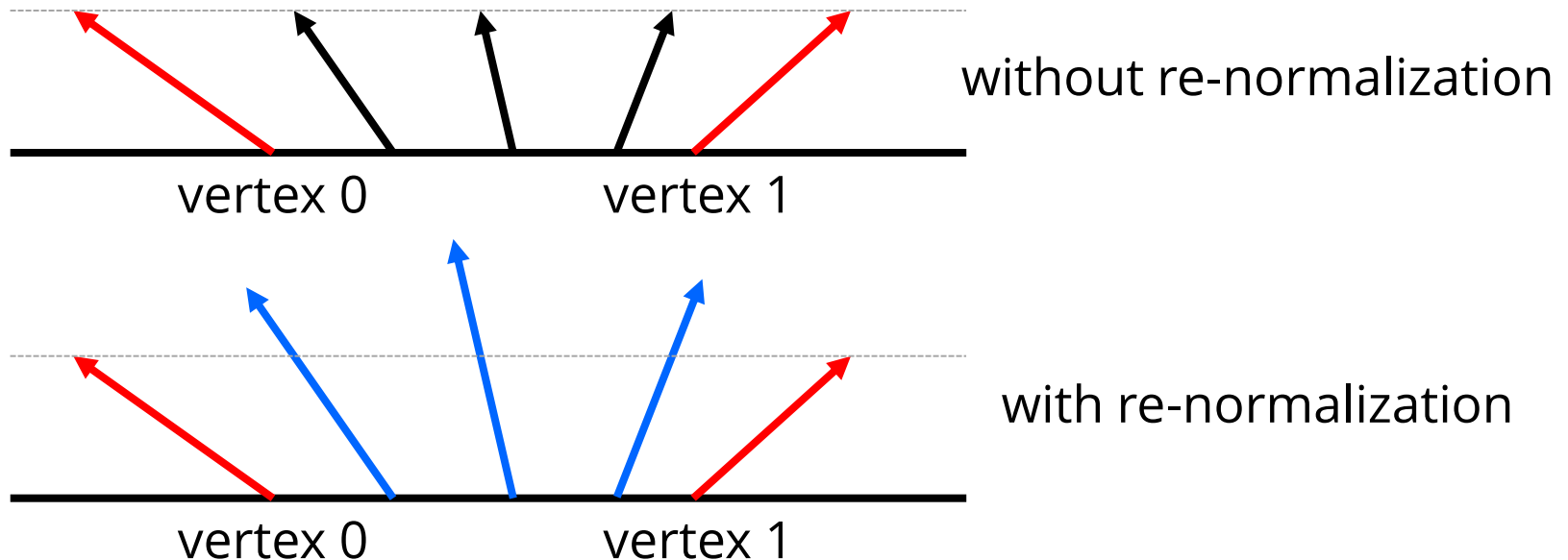
Normals in Phong Shading

- Phong shading requires to re-normalize per-fragment normals.

- Per-fragment normals are linearly interpolated by rasterizer, instead of spherical interpolation.
- So, we need to make them have unit lengths.

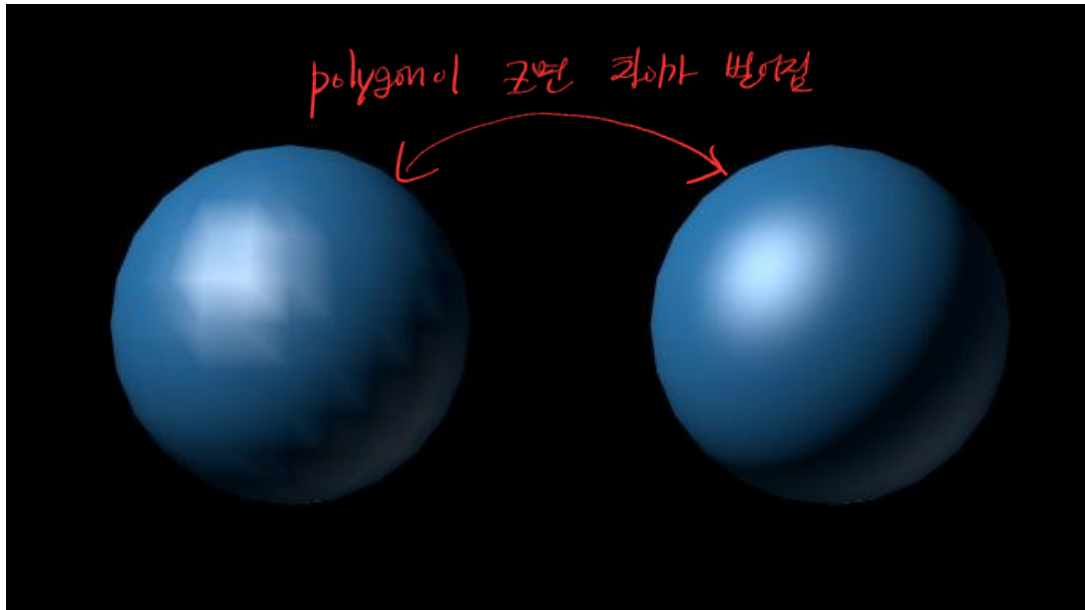
이런 선형적 interpolation

re-normalization



Gouraud vs. Phong Shading

- **Phong shading better preserves details, and much smoother than Gouraud shading.**
 - This difference becomes larger with sharper specular reflections.

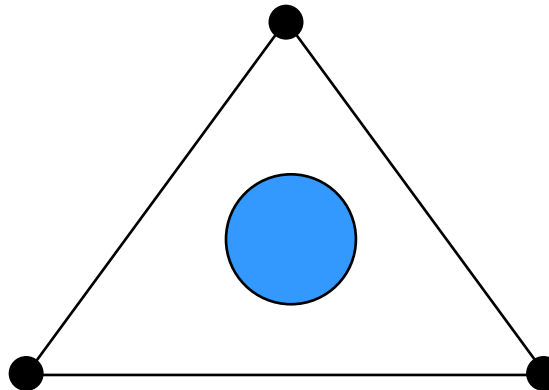


Gouraud vs. Phong Shading

- **Phong shading better preserves details, and much smoother than Gouraud shading.**

- If highlights reside inside triangles (see the figure below), that would be missing in Gouraud shading.
- Example: Blue highlight will be missing in shading of vertices.
- On the other hand, Phong shading interpolates normals, and the per-fragment highlights are captured well.

가장 큰 차이
→ polygon이 카운트를 할 때
각각이 삼각형은 아니



Phong Shading vs. Phong Illumination Model

- **Phong shading** and **Phong illumination model** are different.
 - Phong shading: shading using per-fragment normals
 - Phong illumination model: shading model with ambient, diffuse, specular components.